

Bardiya Alian

[Email](#) | [GitHub](#) | [AstroBin](#)

EDUCATION

Sharif University of Technology (SUT) — *B.Sc. Physics, GPA: 18.10/20* Tehran, Iran, 10/2022–06/2027

AE High School — *Math/Physics Diploma, GPA: 19.57/20* Tehran, Iran, 09/2019–06/2022

RESEARCH EXPERIENCE

Sharif University of Technology (SUT), *Research Assistant to Prof. Reza Rezaei* Fall 2024–Present

Two-Component Dust Emission Analysis of Dwarf Galaxies

- Performing multi-band SED analysis to study the cold and warm dust properties of dwarf galaxies, currently focusing on galaxies in the LITTLE THINGS catalog, and comparing dust maps with star-formation tracers at different wavelengths, including SFR estimates from FUV and NUV data.
- Developing a Python preprocessing pipeline for unit conversion, background subtraction, S/N-based source masking, resolution and pixel-scale matching, and uncertainty estimation through bootstrap resampling.
- Implemented Bayesian/MCMC modified-blackbody fitting and tested the preprocessing and single-component fitting steps on M51. Current work focuses on two-component fitting and dust maps, using dyadic wavelet transforms for larger galaxies.

Star Detection Algorithm for the Iranian National Observatory (INO)

- Developed an automated stellar-source detection method using dyadic wavelet transforms to enhance point sources and suppress background variations, with a focus on crowded and low-S/N images.

Analysis of LITTLE THINGS Galaxies in JHK Bands

- Analyzed near-infrared images for star-formation studies, including missing data, background contamination, and foreground-star removal using J–H color masks. Later shifted the work to submillimeter data because the available JHK observations had limited coverage and quality.

Institute for Research in Fundamental Sciences (IPM), *Research Assistant to Prof. Amin Farhang* Summer 2024

Phase II and Exposure-Time Calculator Development for INO

- Contributed to the development of an ESO-inspired environment for astronomers to prepare INO observation requests. Collaborated with two other interns on scheduling, timing, finding charts, and an initial exposure-time calculator.
- Independently extended the exposure-time calculator to model S/N changes with exposure time, calculate saturation, and identify saturated sources in the field of view, using comparable ESO and Subaru tools as design references.

SELECTED PROJECTS

Deep-Sky Astrophotography [AstroBin](#) Ongoing

- Conducting multi-hour observing sessions involving telescope setup, autoguiding, image acquisition, calibration, stacking, and post-processing with DSLR and equatorial telescope systems.

Non-local Frames in Special Relativity [Document](#) — Special Relativity Fall 2025

- Studied non-local frames in special relativity with Ms. Delaram Dadgoo under the supervision of Prof. Shant Baghram

Gravitational Shock Waves [Document](#) — Special Topics in Gravity Fall 2025

- Studied gravitational shock waves and prepared a report on their theoretical foundations under the supervision of Prof. Bahram Mashhoon.

Torsion in General Relativity [Document](#) — General Relativity I Spring 2025

- Researched and created a presentation on torsion in GR under the supervision of Prof. Sohrab Rahvar.

- CP Violation [Document](#) — Introductory Particle PhysicsFall 2024
 - Prepared a report on CP violation under the supervision of Prof. Amin Faraji Astaneh.
- Speckle Simulation and Adaptive Optics presentation [GitHub](#) — Observational Astronomy, OpticsFall 2024
 - Simulated the effect of atmospheric turbulence on stellar images under the supervision of Prof. Reza Rezaei. Later expanded as a presentation on adaptive optics with Mr. Bardiya Hasanpour under the supervision of Prof. Seyed Nader Seyed Reyhani.
- Spherical Double Pendulum [GitHub](#) — Analytical Mechanics IISpring 2024
 - Simulated the three-dimensional motion of a double pendulum for arbitrary initial conditions under the supervision of Prof. Mahmoud Bahmanabadi.
- Magnetic Mirrors [Document](#) — Electromagnetics IISpring 2024
 - Studied magnetic mirrors and their application to cosmic-ray distributions under the supervision of Prof. Shant Baghran.
- Astronomy Experiment Processing [GitHub](#) — Astronomy LaboratorySpring 2024
 - Completed seven observational experiments in teams of two, involving observations and astronomical image processing.

TEACHING EXPERIENCE

- Optics, *Teaching Assistant to Prof. Seyed Nader Seyed Reyhani*Fall 2026
 - Mentored students, developed assignments, and led problem-solving and review sessions.
- Special Relativity, *Teaching Assistant to Prof. Amin Faraji Astaneh*Spring 2026
 - Mentored students and led problem-solving and review sessions.
- Astronomy Laboratory I, *Teaching Assistant to Prof. Reza Rezaei*Spring 2025, 2026
 - Mentored students during laboratory experiments and held Q&A sessions. Developed and documented a Raspberry Pi HQ Camera system for long-exposure imaging and automated acquisition sequences.

AWARDS AND HONORS

- Silver Medal, Iranian Astronomy and Astrophysics Olympiad
 - Top Astrophotographer, Messier Marathon CompetitionSummer 2021
12/2025

SELECTED COURSEWORK

- Astrophysics I — 19.5/20
 - Astronomy Laboratory I — 19.9/20
 - Observational Astronomy — 19/20
 - General Relativity — 18.4/20
 - General Relativity I (PhD Course) — 16.5/20
- Electrodynamics I — 20/20
 - Quantum Physics I — 19/20
 - Special Topics in Gravity (PhD Course) — 18/20
 - Selected Subjects in Astrophysics — 20/20

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, Matplotlib, Astropy, Photutils, Reproject) LaTeX.

Analysis: Astronomical image processing, multi-wavelength preprocessing, SED fitting, Bayesian inference, MCMC sampling, photometry, source detection, and wavelet analysis.

Observational: Telescope operation, deep-sky imaging, collimation, equatorial mounts, autoguiding, image calibration, stacking, and post-processing.

LANGUAGES AND ADDITIONAL EXPERIENCE

Languages: Persian (Native); English (Fluent; IELTS Academic 8.0: Reading 9.0, Listening 8.5, Writing 6.5, Speaking 7.0); Arabic (Beginner, A1); Latin (Beginner).

Leadership: Cultural Administrator, Physics Scientific Association, Sharif University of Technology (one year); Member, Sharif University Astronomy Club (Spring 2024–Present).

RESEARCH INTERESTS

Dust astrophysics; star formation; galactic structure; observational astronomy; gravity and general relativity.